

**APPROVED**

## IT6010: Maths for Computing

### Course Details

|                    |                                                                                                                       |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| Course Code:       | IT6010                                                                                                                |
| Course Title:      | Maths for Computing <b>APPROVED</b>                                                                                   |
| Short Title:       | Maths for Computing                                                                                                   |
| Course Level::     | Level 6                                                                                                               |
| Valid From::       | 2025/2026 Sem 1                                                                                                       |
| Credits::          | 15                                                                                                                    |
| Owner:             | ICT                                                                                                                   |
| Assessment Method: | Achievement                                                                                                           |
| Course Aim         | To develop the ability to apply standard mathematical and statistical techniques which can be used in the ICT sector. |

| CILO                                                      |                                                                                                         |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| On Completion of this course, the learner will be able to |                                                                                                         |
| #                                                         | Learning Outcome Description                                                                            |
| 1                                                         | Apply appropriate statistical techniques to solve problems in defined and some undefined ICT scenarios. |
| 2                                                         | Use some advanced geometrical methods in simulation applications.                                       |
| 3                                                         | Apply basic and some advanced number theory principles to solve a variety of problems in ICT context.   |
| 4                                                         | Solve cryptography-related problems which are linked to information security.                           |
| Requisites                                                |                                                                                                         |
| No requisites                                             |                                                                                                         |

| Examinations                                   |
|------------------------------------------------|
| <b>Assessment Type</b><br>Examination (Unseen) |
| <b>Assessment Type</b><br>Examination (Unseen) |
| No Other Controlled Assessments                |
| Uncontrolled Assessments                       |
| <b>Assessment Type</b><br>Practical Project    |

| Affiliated Entities |                                                                                        |                |             |
|---------------------|----------------------------------------------------------------------------------------|----------------|-------------|
| Entity Code         | Entity Title                                                                           | Entity Version | Entity Type |
| ICT8011             | Bachelor of Information and Communications Technology (Programming Major)              | 3              | Programme   |
| BFA8001             | Bachelor of Arts in Film and Animation                                                 | 2              | Programme   |
| BFA8800             | Bachelor of Film and Animation                                                         | 4              | Programme   |
| BFA8801             | Bachelor of Arts in Digital Filmmaking                                                 | 3              | Programme   |
| BFA8802             | Bachelor of Arts in Animation and Visual Effects                                       | 3              | Programme   |
| BFT8100             | Bachelor of Science in Financial Technology                                            | 2              | Programme   |
| BFT8101             | Bachelor of Science in Financial Technology                                            | 1              | Programme   |
| ENT7080             | Associate Degree in Engineering Technology (Civil)                                     | 1              | Programme   |
| ENT7081             | Associate Degree in Engineering (Civil)                                                | 1              | Programme   |
| ENT8020             | Bachelor of Engineering Technology (Mechanical)                                        | 3              | Programme   |
| ENT8021             | Bachelor of Engineering (Mechanical)                                                   | 1              | Programme   |
| ENT8025             | Bachelor of Engineering (Industrial Design and Process)                                | 2              | Programme   |
| ENT8031             | Bachelor of Engineering Technology (Electronics)                                       | 4              | Programme   |
| ENT8032             | Bachelor of Engineering (Electronics)                                                  | 1              | Programme   |
| ENT8032             | Bachelor of Engineering (Electronics)                                                  | 1              | Programme   |
| ENT8040             | Bachelor of Engineering Technology (Electrical)                                        | 2              | Programme   |
| ENT8041             | Bachelor of Engineering (Electrical)                                                   | 1              | Programme   |
| ENT8050             | Bachelor of Engineering Technology (Electromechanical)                                 | 2              | Programme   |
| ENT8051             | Bachelor of Engineering (Electromechanical)                                            | 1              | Programme   |
| ENT8060             | Bachelor of Engineering Technology (Communications and Networks)                       | 2              | Programme   |
| ENT8060             | Bachelor of Engineering Technology (Communications and Networks)                       | 3              | Programme   |
| ENT8061             | Bachelor of Engineering (Communications and Networks)                                  | 1              | Programme   |
| ENT8061             | Bachelor of Engineering (Communications and Networks)                                  | 2              | Programme   |
| ENT8070             | Bachelor of Engineering Technology (Chemical and Industrial Processes Engineering)     | 1              | Programme   |
| ENT8071             | Bachelor of Engineering (Chemical and Industrial Processes)                            | 1              | Programme   |
| ENT8071             | Bachelor of Engineering (Chemical and Industrial Processes)                            | 2              | Programme   |
| ENT8080             | Bachelor of Engineering Technology (Civil)                                             | 1              | Programme   |
| ENT8081             | Bachelor of Engineering (Civil)                                                        | 1              | Programme   |
| ENT8090             | Bachelors in Engineering (Industrial Instrumentation and Automatic Control.            | 1              | Programme   |
| ENT8091             | Bachelors in Engineering (Industrial Instrumentation and Automatic Control.            | 1              | Programme   |
| ENT8091             | Bachelors in Engineering (Industrial Instrumentation and Automatic Control.            | 2              | Programme   |
| ENT8100             | Bachelor of Engineering Technology (Biomedical)                                        | 1              | Programme   |
| ICT6010             | Diploma in Information and Communications Technology                                   | 5              | Programme   |
| ICT6010             | Diploma in Information and Communications Technology                                   | 6              | Programme   |
| ICT6025             | Diploma in Cloud Computing                                                             | 2              | Programme   |
| ICT6025             | Diploma in Cloud Computing                                                             | 3              | Programme   |
| ICT6045             | Diploma in Web Development                                                             | 2              | Programme   |
| ICT6045             | Diploma in Web Development                                                             | 3              | Programme   |
| ICT7011             | Associate Degree in Information and Communications Technology (Programming Major)      | 1              | Programme   |
| ICT7022             | Associate Degree in Information and Communications Technology (Information Systems)    | 1              | Programme   |
| ICT7041             | Associate Degree in Information and Communications Technology (Database Systems Major) | 1              | Programme   |
| ICT7051             | Associate Degree in Information and Communications Technology (Cyber Security)         | 1              | Programme   |
| ICT8010             | Bachelor of Information and Communications                                             | 4              | Programme   |

|             |                                                                                              |   |           |
|-------------|----------------------------------------------------------------------------------------------|---|-----------|
|             | Technology (Programming Major)                                                               |   |           |
| ICT8012     | Bachelor of Science in Information and Communications Technology (Programming Major)         | 1 | Programme |
| ICT8020     | Bachelor of Information and Communications Technology (Management Information Systems Major) | 3 | Programme |
| ICT8021     | Bachelor of Information and Communications Technology (Information Systems Major)            | 3 | Programme |
| ICT8022     | Bachelor of Information and Communications Technology (Information Systems Major)            | 3 | Programme |
| ICT8023     | Bachelor of Science in Cloud Computing and Information Systems                               | 3 | Programme |
| ICT8030     | Bachelor of Information and Communications Technology (Networking Major)                     | 5 | Programme |
| ICT8031     | Bachelor of Information and Communications Technology (Networking Major)                     | 4 | Programme |
| ICT8032     | Bachelor of Science in Information and Communications Technology (Networking Major)          | 1 | Programme |
| ICT8040     | Bachelor of Information and Communications Technology (Database Systems Major)               | 5 | Programme |
| ICT8041     | Bachelor of Information and Communications Technology (Database Systems Major)               | 3 | Programme |
| ICT8042     | Bachelor of Science in Data Analytics                                                        | 3 | Programme |
| ICT8044     | Bachelor of Science in Information and Communications Technology (Database Systems Major)    | 1 | Programme |
| ICT8045     | Bachelor of Science in Information and Communications Technology (Database Systems Major)    | 1 | Programme |
| ICT8050     | Bachelor of Information and Communications Technology (Cyber Security Major)                 | 4 | Programme |
| ICT8051     | Bachelor of Information and Communications Technology (Cyber Security Major)                 | 4 | Programme |
| ICT8052     | Bachelor of Science in Information and Communications Technology (Cybersecurity Major)       | 1 | Programme |
| ICT8060     | Bachelor of Science in Artificial Intelligence and Robotics Software                         | 1 | Programme |
| MDM8040     | Minor in Data Mining                                                                         | 1 | Programme |
| No Code Yet | Bachelor of Engineering (Electromechanical)                                                  | 1 | Programme |
| No Code Yet | Bachelor of Engineering (Electromechanical)                                                  | 1 | Programme |
| No Code Yet | Copy Of Bachelor of Information and Communications Technology (Cyber Security Major)         | 1 | Programme |
| No Code Yet | test Bachelor of Information and Communications Technology                                   | 1 | Programme |
| NEDICTE     | Non-faculty Electives                                                                        | 1 | Group     |